Procedural Reconstruction of Slender Plants from Any Field Modality

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Abstract

Editable, biologically interpretable 3D models of field-grown crops support high-throughput phenotyping, in silico *breeding trials, and agronomic digital twins, but recovering such models from field sensor data remains difficult: plant organs are slender, neighboring plants occlude one another, and a field campaign rarely yields more than a single photograph, a noisy depth frame, or a partial point cloud per plant. Slender geometry under occlusion is a regime in which learned 3D completion and Chamfer-based evaluation both fail. In this work, we present* CROPCODE*, a multimodal masked-autoencoder framework that recovers the complete procedural representation of a plant, the generator parameters encoding leaf count, insertion heights, blade angles, and surface geometry, from any subset of RGB, depth, and point-cloud input in a single forward pass. The model is pretrained on 10K–100K procedurally generated maize and sorghum plants rendered with physically simulated row occlusion, and a full-modality teacher is then distilled into single-sensor students. Distillation improves cross-modal generation by 19% on average and parameter recovery from a point cloud alone by 44%. Controlled ablations, replicated across both species, show that the procedural parameters are valuable prediction targets but do not help as a pretraining modality: exposing them to the encoder degrades phenotype recovery on every target we measure. We further show that Chamfer distance and linear probes on the frozen representation rank input modalities differently on slender geometry, and we adopt the probe as the deciding metric. On real field scans with independently reconstructed ground truth,* CROPCODE *matches/approaches the accuracy of a staged annotation pipeline and reduces the cost of procedural reconstruction from hours of manual specification, or minutes of pipeline computation, to milliseconds per plant. We release* FIELDPROC*, the N-plant, two-species benchmark behind these experiments, which pairs multimodal field captures under simulated row occlusion with exact procedural ground truth. These results establish a practical route from fragmentary field sensing to simulation-ready 3D crop field assets at the acre scale.*

Teaser figure: any sensor fragment → one forward pass → complete procedural plant + interpretable parameters

Figure 1. **Any sensor in, procedural plant out.** From any subset of field-practical modalities (a single photograph, a depth frame, or a point cloud truncated by row neighbors), CROPCODE recovers the full procedural representation of the plant: a kilobyte-scale, physiologically interpretable parameter set that re-renders to complete geometry, feeds simulation, and supports trait analytics at field scale. replace box with figure

1. Introduction

Accurate three-dimensional models of individual crop plants enable high-throughput phenotyping of morphological traits such as leaf angle, internode length, and canopy architecture, and they underpin emerging agricultural digital twins that run breeding trials *in silico* [19, 25]. The representation best suited to these uses is procedural: a compact set of biologically interpretable parameters that a generator expands into complete, editable, simulation-ready geometry [20, 35]. A procedural plant occupies a few kilobytes where a laser scan occupies megabytes, every parameter corresponds to a trait an agronomist works with, and the model can be edited, re-rendered, and perturbed in ways that unstructured geometry does not support. Field crops also instantiate a problem of general interest in 3D computer vision: recovering a parametric model of a thin, heavily occluded object from partial multimodal observations.

Yet recovering procedural structure from field-acquired sensor data remains a significant challenge, and every existing route presupposes a rich, clean observation. Manual parameter specification takes hours per plant [20]. Staged reconstruction pipelines automate the process; a recent system annotates laser scans with a vision-language model and fits a procedural descriptor to 5.4 mm median accuracy, but it requires a complete scan of an isolated plant and roughly ten minutes of computation per plant [17]. Per-scene optimization methods search generator parameters against multi-view imagery [52], and recent learned models infer parametric plants directly, but only from a clean, complete point cloud [8, 13]. A phenotyping campaign supplies the opposite: thousands of plants per day, each observed by whichever sensor was practical: a camera, a depth frame, or a partial scan.

A second, geometric barrier compounds the practical one. Crop plants are slender: a maize blade is centimeters wide, millimeters thick, and up to a meter long. In a planted row, each plant is further occluded by its neighbors, whose leaves interleave with the target; in our field simulations, roughly a third of the points inside a plant's bounding region belong to neighboring plants. This regime violates the assumptions of learned 3D completion: the standard benchmarks contain compact, largely convex objects and no slender-under-occlusion category [49, 51], and Chamfer distance, the default metric, is structurally insensitive to thin sheets, so it can reward reconstructions that miss entire blades [42, 47]. Occlusion-aware completion of thin plant organs has begun to address the input side of this regime, but its output is a completed point cloud rather than structure [27]. To our knowledge, no prior method recovers a procedural, re-renderable representation of slender crop organs from a single field-practical observation (a photograph, a depth frame, or a partial point cloud) under row-neighbor occlusion: methods that infer plant parameters require clean, complete scans of isolated plants [8, 13], and methods that tolerate field occlusion complete unstructured geometry instead [27].

In this work, we present CROPCODE, a framework that makes this inverse problem tractable by using the procedural generator itself as the prior. We pretrain a four-stream masked autoencoder [3, 10, 21, 31] over RGB, depth, point clouds, and tokenized procedural parameters of 10K–100K generated maize and sorghum plants, rendered in simulated rows in which neighboring leaves bend on contact. Because random modality subsets are masked during pretraining, a single model learns to complete any subset at inference. A full-modality teacher is then distilled [22] into the single-sensor students that deployment requires. Distillation improves cross-modal generation by 19.1% on average, by 27.5% when conditioning on a point cloud alone, and parameter recovery from that point cloud by 43.8%.

The framework also answers a design question that any model with access to generative structure must face: where should the parameter stream enter? Our controlled ablations give a precise answer: the parameters are valuable prediction targets, and a liability as a representation-learning modality. With token budgets matched, adding the parameter stream to the encoder degrades the frozen representation on every sorghum phenotype target and on most maize targets; the result survives budget-restoring control arms, replicates across both species, and holds even when the parameters serve only as a reconstruction target for the decoder. Supervising parameters as a generation target is nevertheless what makes the distilled generalist outperform a parameter-only specialist at parameter regression itself. One interpretation, consistent with modality competition [34] and negative transfer [48], is that a stream that already contains the answer suppresses learning in the streams that must supply it at test time.

Evaluating these models exposed a metric disagreement with consequences beyond plants. Chamfer distance ranks depth as the most valuable input modality (-32%), whereas a linear probe on the frozen representation, read out on the phenotypic traits the application needs, ranks depth within noise and identifies RGB as the carrier of trait information. On slender geometry the two metrics measure different quantities, surface proximity and recovered structure, and we adopt the probe as the deciding metric for modality decisions [6, 47].

The claim of this work is that fragmentary field sensing suffices to recover a complete, interpretable procedural plant, provided the generator supplies both the training data and the output representation, and its parameters are withheld from the encoder. We validate this claim on real field scans of maize and sorghum with independently reconstructed ground truth [17, 25], including rows in which neighbors occlude the target. one sentence of headline real-data results when available. The key contributions of this work are:

1. **Task and framework.** The first formulation and system, to our knowledge, for recovering procedural plant representations of slender, row-occluded field crops from *any subset* of RGB, depth, and point-cloud input in one amortized forward pass, with full-modal to single-sensor distillation (+19.1% generation, -43.8% parameter error from point clouds alone).
2. **A design rule for structured modalities.** Controlled, budget-matched, two-species evidence that generator parameters are valuable prediction targets, while treating them as an auxiliary representation-learning modality does not improve, and can degrade, the sensor representation; we expect this guidance to transfer to any multimodal model with access to privileged generative structure.

3. **An evaluation protocol for slender geometry.** A demonstration that Chamfer distance and downstream phenotype probes disagree systematically on slender organs, and a frozen-latent probe protocol that resolves modality decisions where Chamfer is blind.
4. **A benchmark.** FIELDPROC, a two-species procedural field benchmark of N plants spanning diversity axes: phenotypes / generator configurations / occlusion severities, with paired multimodal captures, physically simulated row occlusion, and exact parameter ground truth, together with a real-field evaluation split carrying reconstructed procedural ground truth across the genotypes of a diversity panel [25]. No existing dataset combines these elements (Tab. 2) [41, 51]. We will release the data, generation code, and evaluation scripts. confirm release scope and licensing

2. Related Work

Recovering procedural plants. Inverse procedural modeling has a long history for woody plants: parameters of tree generators have been recovered by stochastic search [43], by Markov chain Monte Carlo inference [44], and by inferring L-systems directly from point clouds [16, 28]. These methods optimize or infer per instance. Recent systems address crops specifically. Demeter learns a parametric morphology model from soybean point clouds [8], Ghrer et al. infer grammar-style plant representations from 3D scans [13], CropCraft fits a procedural crop model by Bayesian optimization against multi-view imagery [52], and a recent staged pipeline reconstructs procedural maize from terrestrial laser scans through vision-language-model annotation and differentiable NURBS refinement [9, 17]. Each of these methods presupposes a single rich observation, a clean and complete point cloud or a multi-view capture of an isolated plant, and none addresses missing modalities or inter-plant occlusion.

Programs from observations. Outside agriculture, decoding parametric programs from observations is an established pattern: neural shape programs [45], diffusion-based inverse procedural content generation from a single image [53], text-conditioned CAD generation [24], and cadrille, which maps images, point clouds, or text to executable CAD models [26]. These systems operate on rigid objects observed cleanly and in isolation. Our setting differs in the geometry (deformable, slender organs), in the input (noisy, occluded field sensors), and in a design question that none of them raises: whether the program parameters should be visible to the encoder at all.

Completion and slender geometry. Learned point cloud completion has progressed from folding-based decoders [51] to geometry-aware transformers [5, 49, 50], trained on benchmarks of compact, largely convex objects that contain no slender-under-occlusion category. Completion of thin plant organs under occlusion has emerged recently [27], and reconstruction of thin structures from clean multi-view data is an active topic [12, 38], but in both lines the output is unstructured geometry rather than an editable model. Chamfer distance, the standard completion metric, is insensitive to thin sheets and can be reduced by collapsing predictions onto bulk structure [30, 42, 47]; we therefore evaluate with trait probes alongside geometric metrics (Sec. 4.7).

Multimodal masked pretraining. Masked autoencoding [21] extends to multiple modalities in MultiMAE [3], to any-to-any generation over tokenized modalities in 4M [4, 31], to point clouds in Point-MAE [33], and to combined RGB, depth, and point cloud inputs in EmbodiedMAE, from which our architecture derives [10]. Robustness to missing modalities is a recognized concern [40], and distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks [22]. We extend this machinery in two ways: we add a structured generative modality, encoded with continuous number tokens [14], and we distill a generative full-modality teacher for structured parameter regression.

Learning from procedural data. Pretraining on purely synthetic data is viable, from abstract fractals [23] to procedurally generated scenes [15, 39], with domain randomization bridging the gap to real data [46] and with the value of synthetic data known to depend on the downstream task [11]. Our setting is the favorable case for this strategy: the procedural generator defines the output representation, spans the morphology of the target crop, and is paired with real field evaluation data [25]. Our results add one caveat to the recipe: the generator's parameters belong in the data and in the targets, and should be withheld from the encoder (Sec. 4.6).

3. Method

3.1. Problem formulation

3 short paragraphs from the bullets

3.2. A procedural field-data engine

3.3. Four-stream masked pretraining

2-3 paragraphs

3.4. A vision-complete teacher, single-sensor students

2 paragraphs

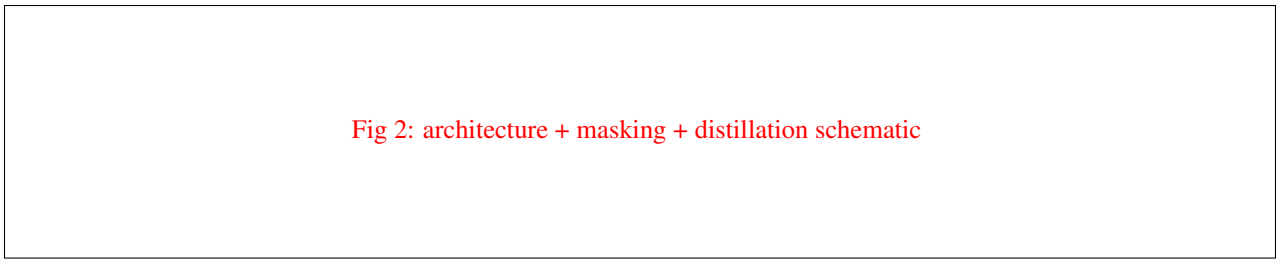


Figure 2. **CROPCODE overview.** (a) Four token streams enter a shared encoder under a Dirichlet masking budget; modality-specific decoders reconstruct all streams. The parameter stream is decoded, never encoded (Sec. 4.6). (b) The full-modality teacher is distilled into single-sensor students for deployment.

	Maize	Sorghum
Plants	N	N
Views / plant	10	10
Parameters / plant	d	d
Leaf count range	3–22	11–27
Occluded-row variants	...	...
Neighbor points in crop	∼30%	...

Table 1. FIELDPROC at a glance. finalize; add diversity axes (phenotypes / generator configs / growth stages) and real-data eval row (MaizeField3D scans + pipeline GT [17, 25])

3.5. Crossing into the field

1-2 paragraphs

4. Experiments

4.1. Setup

compact setup prose; implementation details to supplement

4.2. Regime I: single-view reconstruction of isolated plants

Fig 3 qualitative + edit demo + 2-3 sentences of interpretation

4.3. Regime II: morphological complexity, viewpoint, self-occlusion

Fig 4 + camera schematic + prose

4.4. Regime III: neighbor occlusion in realistic rows

Fig 5 + paired-occlusion table + prose

4.5. Any-subset inference and distillation

4.6. Where the parameter stream belongs

Table (E2 + probe) + prose

4.7. What the metrics disagree about

Fig 6 + prose

4.8. Real-world validation

Table + Fig 7 + prose

4.9. Representation quality and parsimony

Table 5 + Table 6 + prose

5. Discussion and Limitations

5 short paragraphs

6. Conclusion

3 sentences

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	Real	Slender	Row occl.	RGB+D+PC	Proc. GT	Scale
PCN / ShapeNet-55 [49, 51]	×	×	×	×	×	55K
Pheno4D [41]	✓	✓	×	×	×	84
Crops3D [54]	✓	✓	×	×	×	
MaizeField3D [25]	✓	✓	×	×	(✓)	1K
FIELDPROC (ours)	✓†	✓	✓	✓	✓	N

Table 2. Dataset comparison. Proc. GT: exact generative parameters per plant; (✓): fitted post hoc rather than exact. †real-field evaluation split; training data is procedural. verify Crops3D scale; consider adding A-Occ-Plant's data if released

Input	Leaf cnt (acc±1)	Height (rel. err)	Leaf params (mean skill)	F-score ↑	Chamfer ↓
RGB (1 view)					
RGB (3 views)					
Depth (1 view)					
PC (1 view)					
RGB+D+PC					
Two-stage (completion→fit)					

Table 3. **Regime I (isolated plants, maize).** Single-view inputs against the multi-view and multimodal upper bounds; one amortized model against two-stage completion-then-fit. Distilled students throughout; before/after distillation appears in Tab. 4. fill; sorghum replication to supplement

Input	Param err ↓		Geometry ↓	
	before	after distill	before	after
RGB only				
Depth only				
PC only	0.062	0.035		
RGB+D				
RGB+PC				
All three				

Table 4. **Any-subset results, before and after distillation from the vision-complete teacher** (RGB+D+PC; the teacher never observes parameters, Sec. 3.4). Distillation concentrates its gains on the weakest sources. fill; both species

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Procedural Reconstruction of Slender Plants from Any Field Modality

Supplementary Material

A. Detailed Literature Review

This appendix expands the compressed related-work discussion of Section 2 into the five bodies of work the paper builds on or departs from.

A.1. Procedural plant models and inverse procedural modeling

Procedural modeling of plants begins with L-systems [35] and matured into generators whose parameters are organ-level, biologically meaningful quantities [19, 20]. Inverting such models, that is, recovering their parameters from observations, has a long history for woody species: Stava et al. fit tree parameters by stochastic search [43], Talton et al. formulated procedural inversion as MCMC inference [44], and Guo et al. inferred L-systems directly from point clouds [16], with neural variants following [28]. These methods optimize or infer per instance; amortized, feed-forward inversion of a plant generator from field-degraded input is the gap this paper fills. The closest amortized precedents sit outside agriculture: shape-program induction [45], diffusion-based inverse procedural content generation from a single image [53], and cadrille, which maps images, point clouds, or text to executable CAD programs [24, 26]. All operate on rigid, isolated, clean-input objects; none confronts deformable slender organs, inter-plant occlusion, or real-sensor noise, and none studies whether the program parameters should participate in the encoder.

A.2. Learned parametric recovery of crop plants

Two 2025 systems recover parametric plants directly from scans. Demeter learns a morphology model (topology, articulation, PCA shape) from real soybean point clouds [8]; Ghrer et al. infer grammar-style parameterized plants from 3D scans [13]. Both take a single, reasonably complete point cloud as input and neither addresses missing modalities or row occlusion. CropCraft recovers crop structure by Bayesian optimization of a procedural model against multi-view NeRF depth [52]; the approach is accurate but per-scene and slow. A staged pipeline annotates terrestrial laser scans with a vision-language model and fits a NURBS procedural descriptor by differentiable refinement [9, 17, 18]; it handles isolated, occlusion-free plants at $\sim$10 minutes each and serves in this paper as both the occlusion-free baseline and the source of real-data procedural ground truth. Tree-D Fusion reconstructs simulation-ready urban trees from street imagery via diffusion priors [29], but its envelope-then-branch pipeline targets woody silhouettes, a different geometry class from thin cereal blades.

A.3. Point-cloud completion and slender structures

Learned completion runs from PCN [51] through transformer approaches [49, 50] to recent extrapolation–interpolation designs [5], trained and evaluated on benchmarks of compact rigid objects. Neither PCN's eight categories nor ShapeNet-55/34 contains a slender-under-occlusion class, and completion systems assume the missing geometry is self-occluded surface, absent competing clutter from *other* objects inside the crop region. Plant-specific completion is emerging, with amodal segmentation plus completion for occluded plants [27], but it outputs geometry rather than structure or parameters. Curve-aware and thin-structure reconstruction [12, 38] shows growing awareness of the slender regime, on which the standard metric itself fails: Chamfer distance under-weights thin sheets [47], a failure now documented analytically [42] and addressed by density-aware and information-theoretic variants [30, 47]. We take the further step of pairing geometric metrics against frozen-latent phenotype probes and show they can rank modalities oppositely.

A.4. 3D plant phenotyping

Organ-level 3D phenotyping datasets, including Pheno4D's labeled maize and tomato sequences [41], Crops3D [54], and field-scale maize scans with procedural fits [25], anchor trait extraction from point clouds [36, 37]. Radiance-field capture is increasingly practical in field conditions [2]. No existing dataset, however, combines real field capture, slender organs, inter-plant row occlusion, and procedural ground truth; our benchmark is designed to close exactly that conjunction.

A.5. Multimodal masked pretraining and distillation

Masked autoencoding [21] extends naturally to multiple modalities: MultiMAE over RGB, depth, and semantics [3]; 4M and 4M-21 to any-to-any generation over tokenized modalities [4, 31]; Point-MAE to geometry [33]; and EmbodiedMAE, our architectural ancestor, to RGB, depth, and point clouds for robot manipulation [10]. Missing-modality robustness is an active concern [40], and cross-modal distillation from multi-sensor teachers to single-sensor students is established for discriminative tasks; we distill a *generative* full-modality teacher for structured-parameter regression, and encode continuous parameters with xVal-style number tokens [14]. Our finding that a privileged parameter stream suppresses representation quality connects to modality competition [34] and gradient conflict [7, 48], which

supply candidate mechanisms; no prior work, to our knowledge, documents the effect for generator parameters in multimodal pretraining.

A.6. Synthetic-data pretraining and sim-to-real

Pretraining without natural data is viable, from abstract fractals [23] to procedural worlds [15, 39], and domain randomization bridges the reality gap [46], with synthetic value known to be task-dependent [11] and self-consumption a documented risk [1]. Our setting is the favorable case: the deployment target *is* the generator's parameterization, the generator spans the morphology of the crop, and real-field evaluation closes the loop.

B. Detailed Ablation Studies

All numbers in this appendix are from the 15K-plant studies (purple throughout); every table will be refreshed from the 10K–100K reruns. Protocol: every arm trains at a matched budget of 197,400 optimizer steps, global batch 32; representation quality is a ridge probe on the frozen CLS token, one fixed view per plant, identical target vectors across arms; generation quality is validation Chamfer / parameter MAE at the matched checkpoint.

B.1. Modality value-add (E2)

Four pretraining arms differ only in which token streams exist.

Arm	PC Chamfer ↓	vs. PC-only
PC only	0.00495	—
PC + RGB	0.00250	−49.5%
PC + RGB + depth	0.00171	−65.6%
PC + RGB + depth + params	0.00248	−50.0%

Table 5. Sorghum E2: adding the parameter stream to the encoder returns Chamfer to the PC+RGB level, surrendering the depth gain. The frozen probe agrees: the four-modality arm is worse on all eight sorghum phenotype targets and 7 of 11 maize targets. add maize table + probe R^2 columns + CIs

B.2. Token-budget control arms

The four-stream arm shows each vision stream fewer visible tokens, a mechanical confound. Control arms restore RGB/depth/PC to exactly the three-modality arm's 117 visible vision tokens, varying only parameter-text visibility. With the budget restored the gap persists (height probe 0.958 vs. 0.981; roll 0.764 vs. 0.806), and visibility is dose-dependent: at 60% text visible, every skilled target worsens further. The effect is real, and the budget explanation is excluded. full control-arm table with bootstrap CIs

B.3. Parameters as a target only

Masking parameter text at 100% (never an input; reconstructed only by the decoder) still loses to omitting the stream entirely: size targets −0.010 to −0.012 R^2, roll −0.069/ − 0.077 (val/test), mean leaf length −0.102/ −0.097, CIs clear of zero. The latent analysis locates the damage: the target-only arm's latent is the most size-dominated of its cohort (PC1 = size at $|r|$= 0.95, effective rank 4.6). The reconstruction pressure itself, under the encoder's capacity, is what costs structure. reconcile with matched-LR pc2text control when it lands

B.4. Data and model scaling at fixed compute

	Sorghum		Maize	
Train plants	Chamfer	Param acc@.05	Chamfer	Probe moves?
1,000	0.00299	0.608	0.00292*	—
3,000	0.00258	0.775	−34%	yes
10,000	0.00241	0.811	−34%	≤ noise

Table 6. Fixed 197,400-step budget. Near-duplicate 10,000 vs. 10,500 arms give the run-to-run noise floor (∼3% Chamfer, 0.02 R^2). Chamfer keeps paying for data; the probe saturates by ∼3,000 plants in both species. More data buys geometry, and stops buying linearly decodable phenotype. *maize small-data arms actively overfit (best at 29% of schedule, +79% by the end). refresh at 100K; add model-scaling row (25M → 114M = −17.1%; 114M → 332M ≈ 0: base is the knee)

B.5. Distillation, per source modality

Distilling the full-modal teacher into the warm-started student improves the mean generation metric 19.1%, concentrated where the student was weakest: conditioning on PC alone −27.5% (its Chamfer −56.6%), on parameters alone Chamfer −61.9%; RGB and depth, already strong, gain ∼11%. PC→params MAE falls 43.8% (0.0617 → 0.0347), and distilling all four targets beats distilling the parameter head alone (−43.8% vs. −36.9%); the other heads act as auxiliary supervision. Distillation moves the whole view-difficulty curve down without flattening it (9/10 views improve; best-to-worst ratio unchanged), so view robustness needs a different lever. LR caveat: specialist ran at 3e-5 vs. 1e-4; matched-LR control pending. Per-trait recovery tables (PC vs. RGB conditioning, position-only baseline) to follow from reruns.

B.6. View invariance without labels

Ten views of one plant share one parameter ground truth, so the spread of ten predictions is a label-free view-invariance measure. Distillation roughly halves error *and* view spread on the fields with skill (starting point −59%, branching angle −60%); zero-skill fields instead grow *more* view-dependent; the head starts tracking image noise. "Stable"

and "right" are separate axes: several fields move less than 0.3 of their spread across views while remaining worse than predicting the plant mean.

B.7. Frozen-encoder baselines and camera-pose forensics

Against four public frozen encoders under one probe (EmbodiedMAE-B [10], Point-MAE [33], DINOv2 [32], MultiMAE [3]) plus a supervised-from-scratch network, our three-modality arm wins every learnable sorghum target (roll +0.078, mean leaf length +0.104 over the strongest baseline) and the margin comes from fusing RGB with geometry: off-the-shelf Point-MAE beats our PC-only arm, and the four-modality reference arm falls below EmbodiedMAE-B on size targets; this is the parameter-input effect seen from outside. Supplying Point-MAE with each view's full camera pose hands it the generator's fixed plant azimuth, a canonicalization no field rig knows; restricted to gravity alone (what a calibrated rig knows), our pose-free model wins 5 of 9 decided targets. Tassel droop stays below zero for every encoder of this design and remains an open limitation. full tables, both species, with CIs

B.8. Latent structure (E9)

The latents are continua, as they should be for a single-generator population (silhouette ≤ 0.24; k-means at chance against leaf-count deciles); the question is factor alignment. Sorghum latents occupy 4–7 effective dimensions, maize 7–17, consistent with maize's more independent generative factors. The dominant principal component is *never* the phenotype ($\sim$30% of variance unexplained by any trait), and linear decodability dissociates from local embedding structure: roll angle reads at R^2 0.806 from a latent where it is barely locally organized. t-SNE smoothness is the wrong instrument for representation claims on continuous generators; probes are the right one. E9 figures; PC1 identity remains open